HANDLING IMBALANCED AEROSPACE DATA

Predicting Forest Decay using Multi-Spectral Satellite Imagery

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The Ecological Threat

The Problem: “Wilt” is a devastating disease that decays trees from the inside out.

The Global Impact: Over **$30 Billion** in ecological and timber losses annually across affected regions.

The Stealth Factor: Internal rot begins **3 to 6 weeks** before any browning is visible to ground crews.

The Limitation: Traditional ground-based detection is slow, expensive, and only catches the disease in its final stages.

The Aerospace Solution

The Sensor: Quickbird High-Resolution Multispectral Satellite.

The Scale: Analyzing 4,839 forest patches (pixels) simultaneously.

The Invisible Spectrum: Extracting Near-Infrared (NIR) bands to detect biological cellular decay from orbit.

The Goal: Catch the disease *before* it enters the human visual spectrum (Red/Green).

Dataset Overview

The Scale: 4,839 Total Observations (Merged from 4,339 training and 500 testing cases).

The Target Variable: Binary Classification — "Wilt" vs "Healthy" (originally "w" and "n").

5 Predictive Variables:

- *Spectral:* Mean Near-Infrared (NIR), Mean Red, Mean Green.
- *Textural:* GLCM (Gray-Level Co-occurrence), SD_Pan (Panchromatic Standard Deviation).

4,839

Forest Patches Scanned

Data Preparation & Cleaning

Source Tracking: Added 'train/test' origin column to preserve data integrity.

Data Merging: Appended sets in R to create a 4,839-row master file.

Variable Re-coding: Translated raw 'w'/'n' strings to 'Wilt'/'Healthy'.

Factor Conversion: Target converted to an R factor variable for ML modeling.

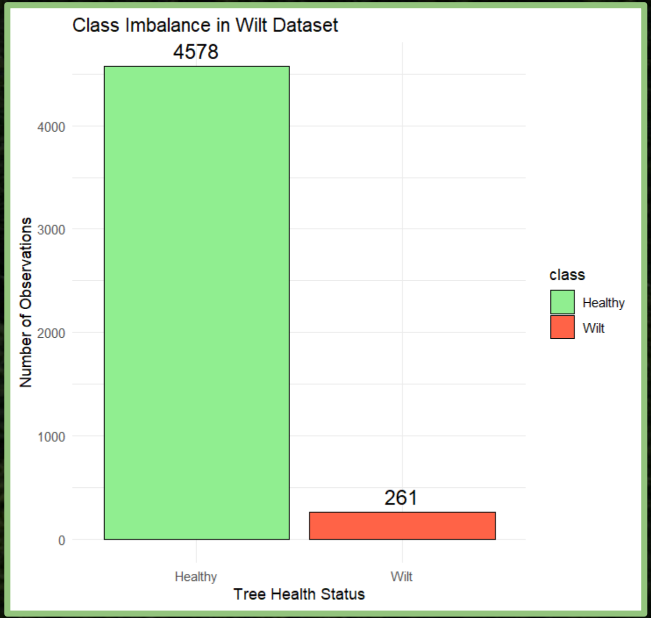
```
> dataset <-
rbind(train, test)

> dataset$class <-
as.factor(dataset$clas
s)

> summary(dataset)
```

The Major Challenge: Class Imbalance

The Reality: **~95%** of the dataset is Healthy. Only **~5%** is Wilt.

The Danger: A naive machine learning model could simply guess "Healthy" every single time and achieve 95% accuracy, completely failing to detect any disease.

The Requirement: We must utilize advanced sampling techniques before building our predictive models to prevent skewed results.

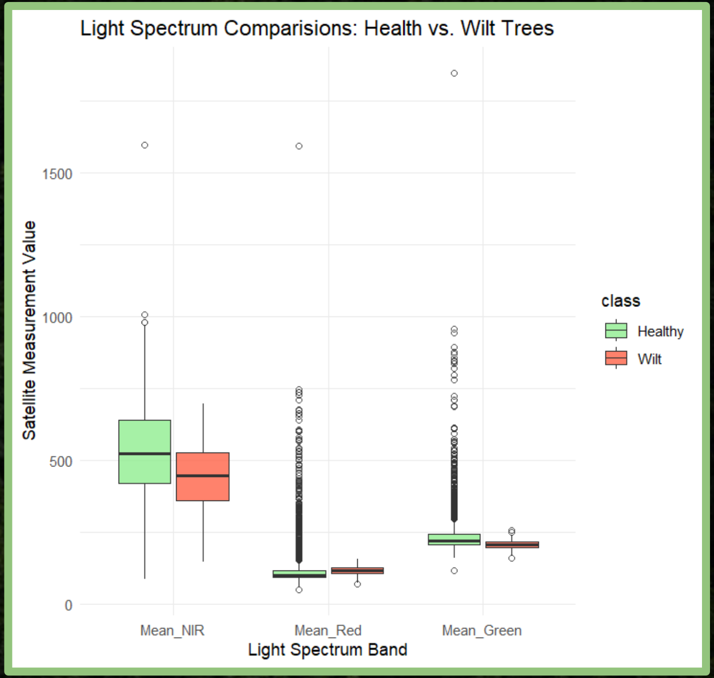


Visualizing Spectral Differences

The Visible Spectrum: Standard **Red** and **Green** light distributions heavily overlap between Healthy and Wilted trees. (This is why ground crews can't see the disease early).

The Invisible Spectrum: Near-Infrared (NIR) shows a distinct, massive drop in diseased trees.

The Biological Reality: As the vascular system of the tree decays from the inside out, its ability to reflect NIR light collapses long before the leaves actually change color.



Proving Differences Mathematically

Mean_Red Light:

- p-value = 0.86 *(Not Significant)*

Mean_Green Light:

- p-value < 2.2e-16 *(Highly Significant)*

Mean_NIR (Near-Infrared):

- p-value < 2.2e-16 *(Highly Significant)*

Result: Red light alone is statistically useless for early detection. The mathematical gap exists purely in the Green and Near-Infrared spectrums.

Confidence Intervals

Metric: 95% Confidence Interval for the difference in population means (Near-Infrared Light)

The Result: [75.25 , 103.82]

The Conclusion: Because this interval **does not cross zero**, we have definitive proof that healthy trees reflect significantly more NIR light than diseased trees. The biological decay is mathematically verifiable.

$$CI = (\bar{x}_1 - \bar{x}_2) \pm t^* \sqrt{[(s_1^2/n_1) + (s_2^2/n_2)]}$$

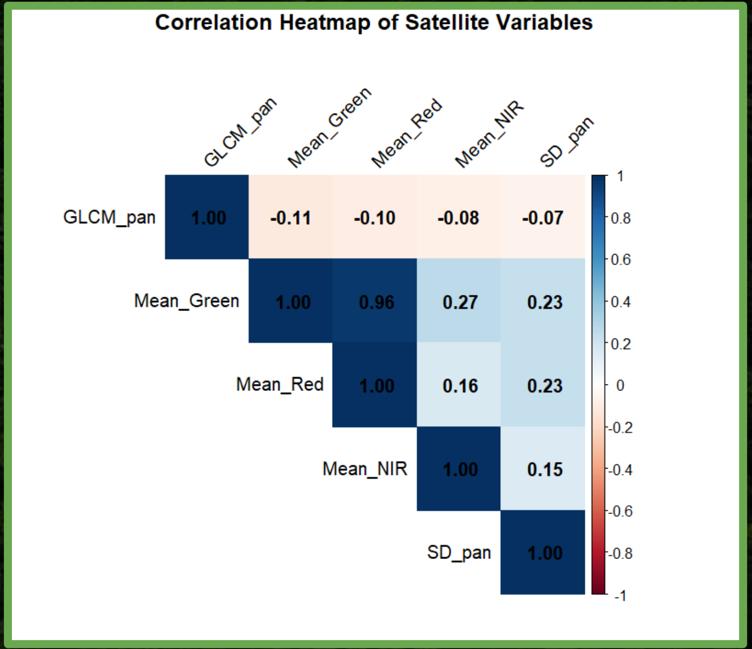
$\bar{x}$ = Sample Mean, s = Standard Deviation, n = Sample Size

Identifying Interacting Variables

Investigating Feature Relationships: Before building predictive models, we must ensure our variables act independently.

The Discovery: We discovered a massive **0.96 positive correlation** between standard Red light and Green light.

The Risk (Multicollinearity): Because Red and Green light move in near-perfect lockstep in this dataset, using both in a model is mathematically redundant and can destabilize our algorithms.



Linear Modeling: Red vs. Green

The Model: `lm(Mean_Red ~ Mean_Green, data = wilt_data)`

The Variance Explained: Multiple R-squared: **0.9309** (93% of the variance in Red light is entirely explained by Green light).

The Slope Confidence Interval: 95% CI: **[0.938 , 0.952]**

The Conclusion: Because the confidence interval is incredibly tight and near 1.0, we have definitive proof we cannot use these variables independently. We must drop Mean_Red from our machine learning model.

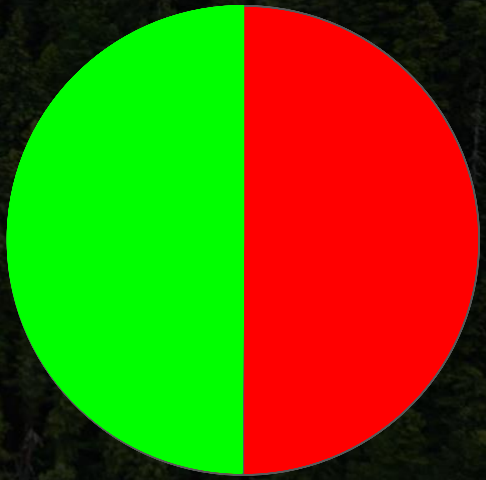
Modeling Prep: Balancing the Data

The Problem: The original training data was highly skewed (4,265 Healthy vs. only 74 Wilt), meaning a model would fail to learn the disease patterns.

The Solution: We addressed this 95/5 imbalance using the `caret::upSample()` algorithm in R.

The Result: New Perfectly Balanced Training Data:

- **4,265 Healthy * 4,265 Wilt**



Building the Random Forest

The Algorithm: Random Forest Classification (`randomForest` package in R).

Feature Selection: Dropped `Mean_Red` entirely based on our linear model findings to prevent multicollinearity.

The Scale: Generated **500 independent decision trees** to vote on the health status of every single forest patch.

```
> set.seed(123)
> rf_model <- randomForest(class ~ .,
                data = balanced_train,
                importance = TRUE,
                ntree = 500)
```

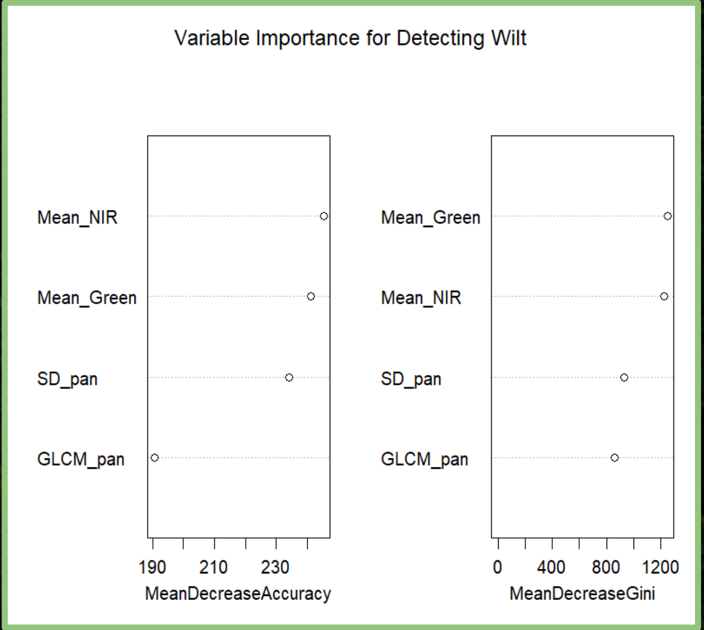
The Wilt Fingerprint: Variable Importance

Our model identified that spectral data is the dominant diagnostic factor in identifying Oak Wilt.

The Top Predictor: Mean_NIR (Near-Infrared) is the #1 most important variable. Its score of **245.3** proves it is the ultimate indicator of cell-structure collapse.

Spectral Support: Mean_Green ranks 2nd (**241.2**), capturing the early chlorophyll breakdown that happens before leaves turn brown.

The Role of Texture: While essential for spatial context, texture variables like **SD_pan** (**234.1**) are secondary to the massive spectral changes in invisible light.



Model Performance & Validation

Overall Accuracy: 63.4% (95% CI: 0.59 – 0.67).

Specificity (Detecting Healthy): 98.4%. The model is nearly perfect at identifying healthy trees.

Sensitivity (Detecting Wilt): 4.8%. The model struggles to positively identify diseased trees in the test set.

The Imbalance Challenge: The extreme disparity between Specificity and Sensitivity highlights the ongoing difficulty of identifying minority classes in real-world environmental data.

	Actual: Healthy	**Actual: Wilt**
Pred: Healthy	308	178
Pred: Wilt	5	9

Analyzing the False Negatives

The Problem: Out of 187 actual Wilt cases in the test set, the model missed 178 of them (False Negatives).

The Investigation: Used `dplyr::filter` to isolate the 178 False-Negative cases from the test set.

The Biological Conclusion: These observations represent **"Stage 1" Early Wilt**.

The Detection Gap: While internal rotting has begun, the canopy color has not yet shifted drastically enough for the satellite sensor to differentiate it from healthy foliage.

The Minimal Variable Model

The Experiment: Dropped texture variables to test a pure "Color-Only" model (NIR + Green).

The Impact: Sensitivity plummeted from **4.8%** to **1.6%**.

The Finding: You cannot drop the texture variables. Even though they ranked lower in importance, all 4 variables are mathematically required.

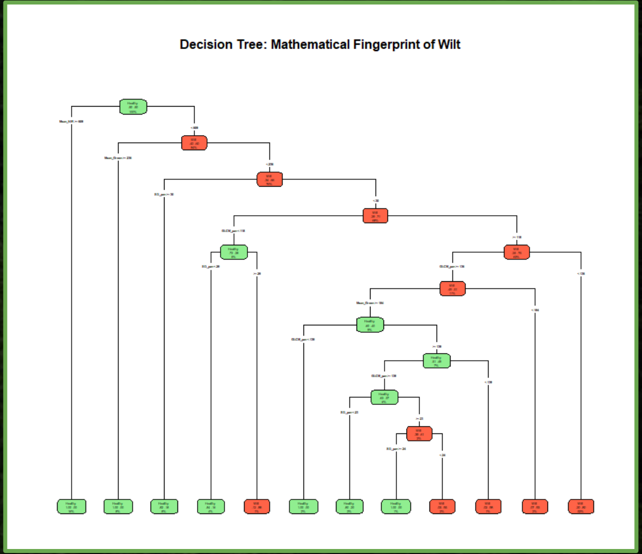
The Mathematical Fingerprint

Interpretable AI: Decoded the Random Forest "black box" using `rpart()` to create a human-readable logic tree.

Exact Thresholds: Identified the precise spectral "tipping points" where a tree transitions from Healthy to Wilt.

Field-Ready Logic: Provides forestry crews with a simple "If/Then" checklist for ground diagnostics.

The Diagnostic Core: Key Trigger: Low **Mean_NIR** + High **SD_pan** = High Wilt Probability.



Final Conclusions

NIR is King: Near-Infrared light is the ultimate predictor of ecological decay.

Interactions Matter: Standard Red light is highly collinear with Green and must be handled carefully.

The Sensor Limitation: Aerospace classification is viable for late-stage wilt, but early-stage detection requires next-generation, higher-sensitivity satellite sensors.

Tools Used: R Studio, Base R, `dplyr`, `ggplot2`, `caret`, `randomForest`.

Reproducibility: The entire analysis is compiled as a dynamic HTML report using **R Markdown (knitR)** for full transparency and reproducibility.

Questions..?

Thank you for your time..